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EXPERIMENTAL RESEARCH IN WHOLE BRAIN EMULATION: THE NEED FOR INNOVATIVE *IN-VIVO* MEASUREMENT TECHNIQUES

RANDAL A. KOENE

*Carboncopies.org & Halcyon Molecular,
505 Penobscot Dr., Redwood City, California 94063, USA
Randal.A.Koene@carboncopies.org*

Whole brain emulation aims to re-implement functions of a mind in another computational substrate by carefully emulating the function of fundamental components, and by copying the connectivity between those components. The precision with which this is done must enable prediction of the natural development of active states. To accomplish this, *in-vivo* measurements at large scale and high resolution are critically important. We propose a set of requirements for these empirical measurements. We then outline general methods leading to acquisition of a structural and functional connectome, and to the characterization of responses at large scale and high resolution. Finally, we describe two new project developments that tackle the problem of functional recording *in-vivo*, namely the “molecular ticker-tape” and the integrated-circuit “Cyborcell”.

Keywords: Whole brain emulation; elements and relations; scope and resolution; empirical data; *in-vivo* measurement; connectome; interfaces.

1. Introduction

The emulation of brain circuitry and in particular whole brain emulation (WBE) are methods that can achieve scientific and medical goals. The scientific utility of a detailed emulation of the neuronal circuitry of a whole brain is as a virtual brain laboratory from which we can learn about the strategies that are employed at different levels of abstraction, while connecting those explicitly with the underlying neurophysiology and neuroanatomy. From a medical perspective, the approach aims at neuroprosthetic technology and ultimately at the ability to transition and transfer the operating functions of a mind onto a computing substrate other than its original biological substrate. In this manner, WBE is a way in which a so-called substrate-independent mind (SIM) can be achieved, a representation of the functions of a specific mind that can be implemented on any computing platform that is able to emulate those functions and thereby elicit the corresponding mental activity.

Carrying out projects towards successful WBE depends both on a formal understanding of that method and on the tools with which to specify its functions and

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populate their parameter values. We introduced a formal conceptualization of the problem in Koene [2012], which we revisit here in Sec. 1.1. The right tools for the task need to be able to acquire structural (connectivity) data and a functional characterization at high resolution and large scale. We will review several of the tools that are already making significant advances toward obtaining the structural data. Because of those advances, and because of the need to make similar strides on the functional side, this paper emphasizes the development of tools with which to acquire functional characterizations at high resolution and at the scale of the whole brain. We present the requirements for whole brain emulation, and we introduce a feasible program of concrete projects that can satisfy those requirements, using present scientific understanding and technology.

1.1. *Matching solid theoretical foundations for whole brain emulation with empirical observations to establish ground truth*

In Koene [2012], we established a general system representation for the effort to achieve whole brain emulation (WBE). This theoretical foundation is purposely technology neutral. Using the notation of optimal control theory, we can express the system energy in terms of a Hamiltonian $H(\mathbf{x}, \lambda, \mathbf{u}, t) = \lambda^T(t)f(\mathbf{x}, \mathbf{u}, t) + L(\mathbf{x}, \mathbf{u}, t)$, with state, $\mathbf{x}(t)$, and costate variables, $\lambda(t)$, being vectors of equal dimension. In Koene [2012], we assumed a representation of mind in terms of a state vector $\mathbf{x}(t)$, with the nonlinear state-space model,

$$\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x}(t), \mathbf{u}(t)) \quad (1)$$

$$\mathbf{y} = \mathbf{h}(t, \mathbf{x}(t), \mathbf{u}(t)), \quad (2)$$

with system input vector $\mathbf{u}(t)$ and system output vector $\mathbf{y}(t)$.

We defined the following set of essential representations for WBE:

- (1) **Mind State**, $\mathbf{x}(t)$: A snapshot of the mental activity that elicits internal awareness and external behavior.
- (2) **Mind Transition Functions**, $\mathbf{f}(t, \mathbf{x}(t), \mathbf{u}(t))$: Functions that describe the dynamic development of the Mind State, making it possible to predict (with acceptable deviation, $\epsilon < \epsilon_{\max}$, and for a restricted time interval, $\Delta t < \Delta t_{\max}$) what a Mind State $\mathbf{x}(t + \Delta t)$ should be, given a known Mind State $\mathbf{x}(t)$.
- (3) **Transition Update Functions**, $\mathbf{g}(t, f(t), \mathbf{x}(t), \mathbf{u}(t))$: Describing the changes of Mind Transition Functions.

To implement an emulation, we must identify sets of elements of the neurophysiology and their relations in $\mathbf{x}(t)$, $\mathbf{f}(t, \mathbf{x}(t), \mathbf{u}(t))$ and $\mathbf{g}(t, f(t), \mathbf{x}(t), \mathbf{u}(t))$, and we must measure corresponding parameter values. *In-vivo* experimental measurements are essential. To get all the necessary data for WBE we need to do two things:

- (1) Establish the well-constrained set of parameters that characterize $\mathbf{x}(t)$, $\mathbf{f}(t)$ and $\mathbf{g}(t)$ at sufficient scope and resolution [Bialek *et al.*, 1991]. This involves carrying

out experiments that test our hypotheses in this regard. An example is the work by Briggman *et al.* [2011] that tests the correlation between the observed structure of circuits of cells in retina and their characteristic function. Another example is work by David Dalrymple that aims to emulate neural circuit function in *Elegans* from a functional analysis without directly acquiring data at the molecular level.

- (2) Acquire the complete set of parameter data at that scope and resolution. To do this we need to replace or improve our measurement tools so that they are adequate for the task. Projects in that regard are described in the following sections.

For the first point, we need to be able to compute a deviation $\epsilon_{\max}$ when carrying out an emulation at a chosen scope and resolution for a specific time difference Δt . Notice that there is some freedom of choice here. There are two important emulation choices in particular: (a) What is the desired time difference Δt_{emu} for which we wish to guarantee a correspondence of the dynamic development of the emulation? (b) What is the greatest deviation $\epsilon_{\max, \text{emu}}$ that we consider acceptable for the emulation, the threshold value below which we consider the emulation sufficiently precise? These questions involve an element of choice, a consideration of what is adequate. Answering them must therefore include the consideration of specific goals, which is a topic large enough to deserve one or more of its own dedicated papers. We will publish such a treatment of Δt_{emu} and $\epsilon_{\max, \text{emu}}$ separately, and so we will here defer to that publication.

When we have a set of target values, Δt_{emu} and $\epsilon_{\max, \text{emu}}$, then the actual $\epsilon_{\max}$ obtained for an emulation over a time step $\Delta t = \Delta t_{\text{emu}}$ tells us if the scope and resolution used in representations $\mathbf{x}(t)$, $\mathbf{f}(t)$ and $\mathbf{g}(t)$, and the parameter values set in accordance with data acquired from the brain satisfy the goals of the WBE in terms of the precision threshold $\epsilon_{\max, \text{emu}}$. For small Δt and small networks or networks with highly constrained propagation over the interval Δt , it may be feasible to determine or estimate $\epsilon_{\max}$ theoretically with knowledge of the limits of deviations that are highly localized in space and time. Each time when we opt to constrain scope or resolution in a specific way, our component models may give us some knowledge of those limits. Beyond that, with an implemented emulation, there are numerical sampling protocols that can be used to estimate $\epsilon_{\max}$. One such protocol is the following:

- (1) Obtain a set of measurements in a biological brain at time t . Record another set of measurements at time $t + \Delta t$.
- (2) Initialize the emulation with parameter values for $\mathbf{x}(t)$, $\mathbf{f}(t)$ and $\mathbf{g}(t)$ based on the measurements at t .
- (3) Carry out emulation from $\mathbf{x}(t)$ over the time interval Δt . Note the resulting $\mathbf{x}(t + \Delta t)$.
- (4) Compare $\mathbf{x}(t + \Delta t)$ with the parameter values that would be set by using the observed data at $t + \Delta t$. Using an agreed-upon metric for the difference, we obtain ϵ .

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Note that this is also possible with further abstraction by using only information about $\mathbf{u}(t)$ and $\mathbf{y}(t)$, instead of $\mathbf{x}(t)$. That is a black-box treatment, as described in Sec. 2.3.

If we consider all things that could belong to our scope then we can draw up a truth table with all the ways in which the data scope can be populated. We can do the same for resolution by listing all of the possible contributing effectors to include. Together (e.g., Table 1), the possible combinations span the space of possible hypotheses for the types of emulations that can be created, some of which satisfy $\epsilon_{\max, \text{emu}}$ and some of which do not. Of course, there are considerably smarter ways to explore this space than to try all possible combinations.

It is likely that there are physical limitations that can help us constrain scope and resolution intelligently. The variance caused by system noise affects the precision with which responses can be predicted even with identical system state and input. For example, if we consider a small sub-net of neurons, the times of their spike responses will not be identical in multiple instances where the unchanged sub-net receives the same stimulation. This limitation of the computational precision provides insight about operational constraints on the sub-net of neurons.

Similarly, there are task-specific limits, some of which depend on physical characteristics of the environment, of physical sensors and of motor actuators. Those limits constrain the precision, the rate and the quantity of information processing in a given task. The brain systems that are involved with carrying out that task need only satisfy those constraints. If we have good insight into the range of tasks, and the corresponding constraints that some brain system (e.g., a region or network of neurons) is involved with, then we can determine their maxima. Scope and resolution of an emulation need to be great enough to address that range. It can differ from system to system. Notice that what we consider essential capabilities, tasks that the system must be able to carry out, is again related to choices about target values Δt_{emu} and $\epsilon_{\max, \text{emu}}$. Perhaps, the system should be able to identify regular features of real-world objects in visual input, or perhaps it should exhibit a different response depending on whether the ambient system temperature is elevated or lowered [Aho *et al.*, 1993].

As the temperature example in the previous paragraph already indicates, it is important to realize which physical signals are of interest, and therefore what data needs to be acquired from the biological brain. It is very likely that we are interested in spikes of the neuronal membrane potential. Perhaps we are also interested in the shape of the membrane response, in conductance through receptor channels, the rate at which diffuse messengers arrive, or changes of local temperature. When we have selected the observations we wish to make, then we can implement a measurement solution in steps: (a) Make measurements, (b) transform measurements into a form suitable for delivery, (c) deliver measurements to be recorded. Preceding the recording, there may be a data aggregation stage.

It may not be necessary to make direct measurements in order to give values to each of the parameters of $\mathbf{x}(t)$, $\mathbf{f}(t)$ and $\mathbf{g}(t)$ in an emulation. A good characterization

Table 1. An example showing a limited subset of the scope and resolution decisions that can be made.

Scope										Resolution			
CNS				PNS		Embodied		Environment feedback	Neurons			Glia Modulation	
Prosencephalon	Brain stem	Spinal cord	Somatic	Autonomic	Senses	Musculature	Spikes		Morphology	Receptors			
T	F	F	F	F	F	F	F	F	T	F	F	F	
T	T	F	F	F	F	F	F	F	T	F	F	F	
T	T	F	F	F	T	T	T	T	T	F	T	F	
T	T	F	F	F	T	T	T	T	T	T	T	T	
T	T	F	F	F	T	T	T	T	T	T	T	T	
T	T	T	F	F	T	T	T	T	T	T	T	T	
T	T	T	T	T	T	T	T	T	T	T	T	T	
T	T	T	T	T	T	T	T	T	T	T	T	T	

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of typical components of a brain enables the construction of catalogs in which components are categorized. Each of the categories can describe a correspondence between one or a few measurements and the probable specifications of parameters within distributions that apply to that category. With adequate validation, error estimation and error correction, the dimensionality of the data acquisition problem can be greatly reduced in this manner. For example, we may infer a category and receptor arrangement for a neuron from a measurement of its characteristic voltage response. Another example is the ability to map morphology, as obtained through techniques in structural connectomics [Hayworth, 2012], to types of neurons and their functional involvement. Such a mapping has been the focus of recent investigations [Briggman *et al.*, 2011; Bock *et al.*, 2011].

2. New Approaches to Large-Scale High-Resolution Measurements

Our ability to acquire the necessary data for a WBE hinges on having the right tools. We can improve existing tools or we can develop new ones that use strategies aimed specifically at the data type, resolution and quantity required. We know that these tools are feasible, because nature has already shown us that data can be acquired, used, transformed and transmitted at the requisite scale and resolution, namely through its implementation of processes that are sensitive and interconnected at that level within the brain itself. Our task is to find the most efficient route to those capabilities with and from the technology that is available to us today.

New measurement tools at large scale and high resolution should meet a number of performance requirements, and we list many of those here in order of criticality:

- Very large scale neural recording or interfacing. Recording very many data points during one session of concurrent measurements.
- High spatial or component resolution during data acquisition. Recording measurements at a parameter resolution that makes emulations possible that satisfy $\epsilon_{\max, \text{emu}}$.
- Recording or interfacing sites distributed throughout the brain, i.e., brain-wide scans. The scope of data acquisition must meet the demands for a satisfactory emulation.
- High rate of data acquisition. Recording measurements at a temporal resolution that makes emulations possible that satisfy $\epsilon_{\max, \text{emu}}$.
- Precise and reliable absolute or relative location registration for each measurement site. This facilitates the alignment of functional characterizations with the connection structure.
- Reliability of the interface; reliability of the scanning technology. The greater the reliability of measurements, the fewer measurements need to be taken and the less validation work needs to be done.
- Avoid damage to healthy tissue. Less damage not only means better prospects for *in-vivo* procedures, but also means better resulting data.

- Low impact on function and activity. A method that does not impair normal function may be usable *in-vivo*. It also reduces the influence that the measurement process may be having on the data being acquired, which is something we need to take into consideration due to the plastic nature of the brain.
- Safety of the interface implantation procedure, or safety of the (functional) scanning procedure. Low-risk methods of application will increase the rate of successful recordings. By presenting a lower hurdle, they will also accelerate the frequency of application and thereby the rate of improvements.

A conceptual approach that helps with several of the most critical requirements is the aggregation of data *in-situ*. We therefore find an aggregation phase in a number of the new tools being developed. Another common feature is wireless collection of data, which has non-intrusiveness, scale and bandwidth benefits that are understood in external imaging, but that apply equally to internal methods of data collection. Hybrid approaches utilize internal measurement, *in-situ* aggregation of data, and wireless or passive transmission to a large-scale scanning or imaging mode. In its simplest form, the internal part of the sequence involves contrast agents that improve the visibility of otherwise hard to detect signals.

2.1. Power, placement and bio-compatibility

Distributed interfaces and measurement devices will require either the distribution of power that they can receive wirelessly or an *in-situ* means of harvesting energy. When the device implementation is biological (e.g., see “molecular ticker-tape” below) then chemical processes harvest energy. In non-biological implementations, it is not immediately clear which of the two approaches is more advantageous. Harvesting energy *in-situ*, such as from glucose, for use in devices is an active area of research and development. Glucose biofuel cells have been shown to work continuously for months within rats [Cinquin *et al.*, 2010]. We may even consider a combination of energy sources, receiving some through transmissions from external sources and some *in-situ*.

Large-scale interfacing cannot be achieved by “manual” placement of the sensor network. So, instead, physical implementation of a functional neural interface may rely on processes such as: Diffuse or random distribution, a self-organizing spatial hierarchy, or taking advantage of existing currents that lead to desired targets (e.g., blood in brain capillaries, 3–4 times daily replacement of cerebrospinal fluid). The deployment of specific physical shapes at strategic moments can influence distribution and positioning. There are, for example, schemes for the autonomous creation of shapes through folding DNA [Rothermund, 2006]. Processes of that sort can be triggered by signals such as neural activity, and they can therefore take part in targeted positioning.

Using *in-situ* devices for data acquisition means having to take into consideration biological compatibility and avoiding adverse effects. It may be useful to deliver biological stimulants, chemical stimulants or nanomaterials to recording sites before (“priming” the site) or during the data acquisition process, for example, to insure that

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regulatory networks within neuronal cell bodies react appropriately to the interface. In some cases, this involves creating an interface channel, as demonstrated by neurotrophic electrodes [Kennedy *et al.*, 2000; Bartels *et al.*, 2008]. Active modifications of interface sites that require adaptation to the interface is useful in the case of the neurotrophic electrode used in work at selected sites by Kennedy *et al.* But note that this can have drastic consequences if done at millions of neurons throughout the brain. Such large-scale adaptation could disrupt normal functions to the point of fundamentally altering mind functions. Procedures that avoid adverse effects include the following measurement interfaces that are essentially undetected by the neuronal network: (1) Biologically inert sensing probes. (2) Probes that become part of the signaling circuit. They may divert some signal flow, but provide the same expected output.

2.2. *Methods of measurement and aggregation*

The general problems are: (A) To carry out measurements at the desired scale and resolution. (B) To detect and collect data about the signals that are of interest. To deal with the first problem, the measurement technique must be a good match to the data resolution believed necessary, while preferred scale is the whole brain. Since the resolution question is unresolved at this time, we will address remaining requirements first and assume here that we can treat individual neurons as information processors and producers. One way to deal with problem B is to divide the problem into multiple steps:

- (1) First, we focus on detection at the cellular scale. The biological substrate uses means to detect or react to relevant signals at sufficiently high resolution. The range of mechanisms used in that substrate can inspire our measurement procedures. Contrast dyes, voltage- and calcium-sensitive dyes, and q-dots (see below), are the precursors of this strategy.
- (2) Once we can carry out the measurements, the next steps are to make necessary conversions and carry out delivery of the measurement in a manner well suited to our data aggregation method.

As we consider many separate tasks (detection, conversion for delivery, delivery, conversion for aggregation, etc.), we may also consider many specialized agents. The complete task of data acquisition is then a coordination of activities.

2.2.1. *Existing in-situ signal enhancers and detectors*

One of the most popular means of acquiring system-wide dynamic data is the use of voltage sensitive dyes, for which the dynamic range is being improved [Kralj *et al.*, 2011]. These are the immediate successors in many applications of the prevalent fluorescent calcium imaging methods, because changes in calcium concentrations are comparatively slow. On the stimulation side, the development of optogenetics [Boyden *et al.*, 2005; Deisseroth, 2011] has made it possible to stimulate in a highly selective manner and to observe clearly correlated downstream effects.

Other efforts of the last decade have resulted in contrast agents for use with magnetic resonance brain imaging (MRI) and interoperative optical imaging [Kircher *et al.*, 2003], and have combined imaging and treatment possibilities such as photodynamic therapy, in which the uptake of a sensitizer by cancer cells is followed by photoirradiation to activate the sensitizer [Reddy *et al.*, 2006]. A family of contrast agents based on superparamagnetic iron oxide nanoparticles and calmodulin have been tested as a means of visualizing levels of intracellular calcium, in effect enabling functional molecular imaging of biological signaling networks in live, opaque specimens [Atanasiyevic *et al.*, 2006]. A promising semiconductor development, quantum dots (or q-dots), may replace the use of some organic dyes [Walling *et al.*, 2009].

Quantum dots are a semiconductor nanostructure (e.g., cadmium selenide, CdSe) with conductances determined by the size and shape of the individual crystal [Reed *et al.*, 1988]. Depending on the fabrication process (self-assembly or etching and lithography), dimensions are between 5–10 nm and 50–100 nm. Q-dots can be functionalized to bind to specific sites, acting as nanoprobe to interact with individual cells. For example, this is presently done to bind them to tumor sites, but the same principle may enable targeting of synapses. Imaging with q-dots can be done for extended periods (months) due to much greater photostability of the q-dots compared with organic dyes. This stability also makes it possible to use q-dots in order to track the movement of individual molecules and cells.

Despite these advantages, q-dots have yet to receive approval for application in humans. Uncoated q-dots are highly toxic under UV radiation. Polymer coated q-dots appear to be non-toxic, but further examination is needed, for example, to determine the excretion process of q-dots.

2.2.2. Aggregation

Aggregation of the collected data may be accomplished by transmission via brain imaging, by on-agent storage and recovery of those agents, or by an agent-to-agent communication network.

The diagrams in Fig. 1 suggest two variations of the concept of aggregation by brain imaging. In Fig. 1(a), we envision an active transmission agent, which may use as its power source electromagnetic radiation that is provided externally. The agent converts measurements of an interesting biophysical signal into an emission that is well-suited to imaging at high spatial and temporal resolution. The imaging technology used does not need to conform to the constraints of current technology, since *agent emissions and their detection can be optimized for each other*.

Figure 1(b) depicts the equivalent setup with a passive agent. Measurement of a relevant biophysical signal causes changes in the properties of the agent, and that affects the interaction of that agent with an imaging scan. In the example depicted, the agents become reflective when modified following biophysical measurement, though other modulations are possible.

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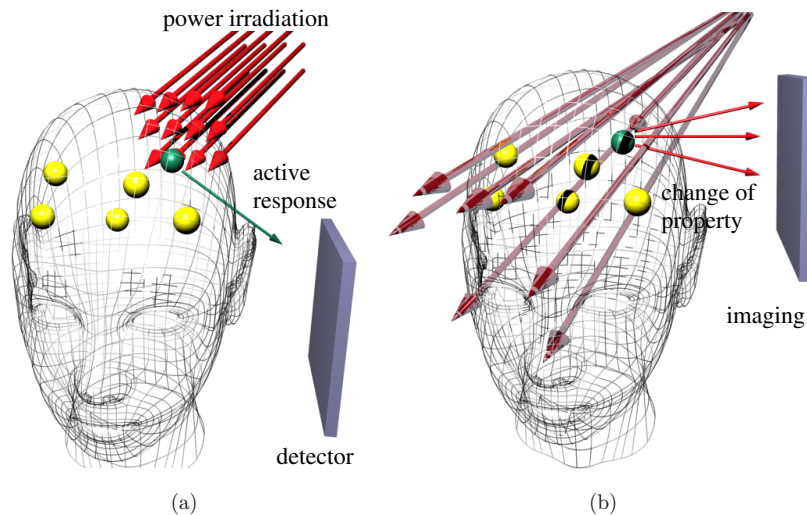


Fig. 1. Concepts for *in-vivo* agent-enhanced aggregation of large-scale high-resolution data. In (a) agents are powered by electromagnetic radiation and the detection of a relevant biological signal causes an active response (green). In (b) passive agents change properties in response to biological signals, which changes their interaction with imaging transmissions (see, e.g., micro-OPID in Sec. 3.2).

2.2.3. Measurement registration

Important information about any measurement are time and place of the measurement. We may substitute relative for absolute time and location indications if necessary. For example, we may register the order of measurements and the positions of agents in relation to one another. In the case of aggregation by brain imaging, this

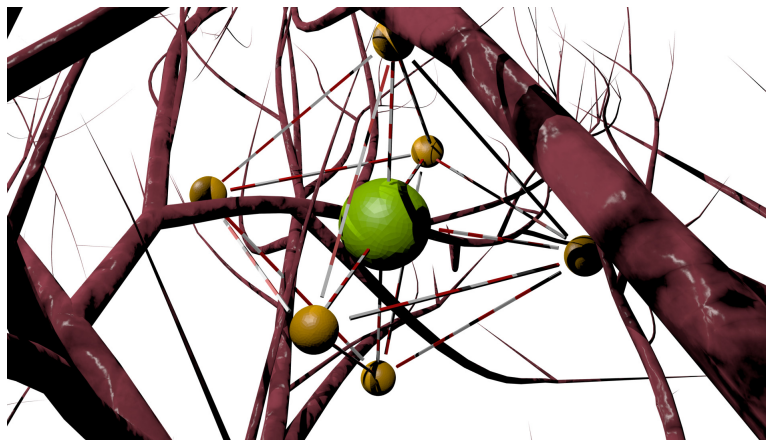


Fig. 2. Relative location information is frequently sufficient. Data collection agents (orange) can establish their location with respect to each other and hubs (green). Hubs may specialize in logistics and infrastructure tasks, such as transmitting locations. A complete map of location registrations can then be obtained *in-vivo*.

information may be obtained as part of the imaging scan. Alternatively, determining and providing this information may be one of the tasks carried out by sets of agents. For example, relative location information between agents may be determined through passive or active properties of the agents, which may then be conveyed by an agent-to-agent transmission protocol.

What we really want to register are associated sets of information about the brain: Information about the structure of connected components, information about the functional characteristics of components, the time at which measurements were taken, the locations within the brain at which measurements were made and their relationships to structures and components. Note the utility of multi-scale scans in order to verify inferred parameters, establishing if proposed models fit observations at different scales.

2.3. Structural connectomics, functional connectomics and large-scale high-resolution functional characterization

Representation levels: Whenever we are trying to describe a process, for example by building models, we can choose the level of representation that we think best suits our needs. In computational modeling, as throughout neuroscience and cognitive science, these choices are the first step that deserves careful review and justification.

At and below the chosen level, one treats components as black boxes that transform input into output in a certain way. A good functional characterization of the components is critically important. If we choose a high resolution then the advantage is that components are simpler. Their greater proximity to the basic physics helps us capture all of the relevant behavior. A disadvantage is that there is a greater challenge in terms of the physical process of data acquisition at such resolutions and in handling large amounts of data. If we choose a lower resolution then the components are more abstract and capturing all of their processing behavior will require more observations over an extended period of time. There is then a greater risk that latent functions are missed.

Above the chosen level, one needs to understand the way in which components interact. A sufficiently complete and correct structural characterization is essential. The connectome must be obtained, because interactions produce emergent system behavior. It is possible to make pragmatic trade-offs. For example, if we identify components at a lower resolution (e.g., whole neurons instead of the morphology of neuronal arbor), but functionally characterize many of them concurrently, so that we can observe covariations in the patterns of activity, then it is possible to infer a functional connectivity structure. The lower resolution does mean that we need to be concerned with missing latent function, which is especially problematic when components are silent.

We could introduce stimulation to put the system through its paces, but that still requires an extensive period of observation and in a plastic system such as the brain, that introduces the possibility that our measurement procedure causes changes.

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Alternatively, we can attempt to reduce the types of measurements that need to be made. For example, we might measure only structure at a high resolution and rely on a mapping from reconstructed component morphology to a library of characteristic morphologies and distributions of parameter values. In that way, we may be able to identify and reimplement component function using the mapped morphology. Of course, that only works if the mapping is one-to-one.

It is also important to consider measurement error. Neuronal networks are known to be robust to random errors, but if our measurements contain non-random errors (e.g., size measurements that tend to err in one direction) then the cumulative error may not be noticed or corrected without additional types of measurements. Validation is a serious issue, as there will always be some error. It is important that the system of combined components makes sense at a higher level. Functional and structural measurements together lead to a more reliable result, and from an engineering perspective we would prefer to test partial reimplementations *in-situ* to validate their satisfactory operation. Whatever the chosen resolution, any complex system (e.g., human, computer, etc.) has characteristic functions and characteristic structure. In Fig. 3, we see common representation levels chosen by different approaches to substrate-independent minds (SIM).

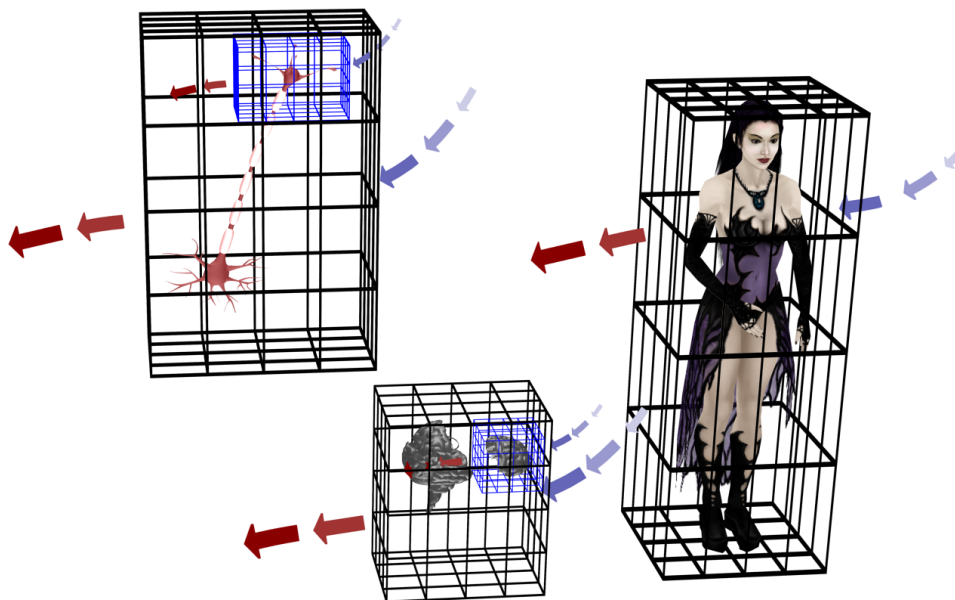


Fig. 3. Examples of “black-box” levels for model representations. Within the black wire-frames we see the whole neuron level, the whole brain level and the sensori-motor body-behavior level. Within blue wire-frames we see parts described within a morphological neuron level and within a brain module level. Arrows indicate input (blue) to processes at a chosen level of representation and output produced (red). Data collected has to give us the actual structure of interaction between components at the chosen level and above. The data must also provide enough characteristic information about input–output relationships at the chosen component level to enable a sufficiently precise model of the transfer function that each component represents.

Technologies: Sets of different technologies work at different levels of black-box representation. The following is not an exhaustive list. At the molecular, neurotransmission channel, neural morphological and neural activity levels we find methods of characterizing component function at high resolution and large scale, and of obtaining a structural or functional connectome. This will be described in more detail below and involves technologies identified as: volume microscopy, carried out by serial block-face scanning electron microscopy (SBFSEM), automatic tape-collecting lathe ultramicrotome (ATLUM), focused ion-beam scanning electron microscopy (FIBSEM), or the knife-edge scanning microscope (KESM) [McCormick and Mayerich, 2004]; the “molecular ticker-tape”; “Cyborcells” based on integrated circuit technology; a “Demux-Tree” that may be composed of nanowires [Watanabe *et al.*, 2009]; neural probes based on electrodes; and optogenetic manipulation [Boyden *et al.*, 2005].

At the neural population, brain module and cognitive system levels we find: electroencephalography (EEG); functional magnetic resonance imaging (fMRI); transcranial magnetic stimulation (TMS); and proposed protocols to tune standardized cognitive architectures to express individual characteristics.

Going from the cognitive system level to the full body or brain sensori-motor and behavioral levels we find: augmented reality devices; virtual reality; systems that use machine learning to acquire your characteristic behavior; and the Bainbridge–Rothblatt inventory for data collection [Bainbridge, 2006].

Goals and categories: The objective of substrate-independent minds (SIM) involves two major goals, namely the ability to reimplement functions of the mind in another computing substrate and, through improved access, to expand the mind’s capabilities, an interface with new data and processing opportunities. It is because of those two goals that scientific insight and technology development in both of those directions are considered to contribute significantly to SIM.

In discussing technological approaches, we use three major category terms that serve mostly to constrain the levels at which applicable technologies should operate. Shown in Fig. 4, those terms are whole brain emulation (WBE), brain–computer interfaces (BCI) and loosely-coupled off-loading (LCOL). The pragmatic philosophy behind whole brain emulation is that we do not understand enough about the wealth of functions that the mind can carry out, the strategies used, or the hierarchical composition of those strategies. We do know quite a bit about the types of functions that are carried out by more basic components of the brain’s physiology, so that if we reimplement at that level of our understanding then we are much more likely to achieve the intended result. Consequently, researchers in WBE typically constrain levels of representation to neural activity, neuron morphology and connectivity, neural transmission channels and molecular processes.

Loosely-coupled off-loading approaches take an almost opposite stance, focusing on the data acquisition technology that is available without further development, which can be best applied top–down at the highest levels (lowest resolution) of

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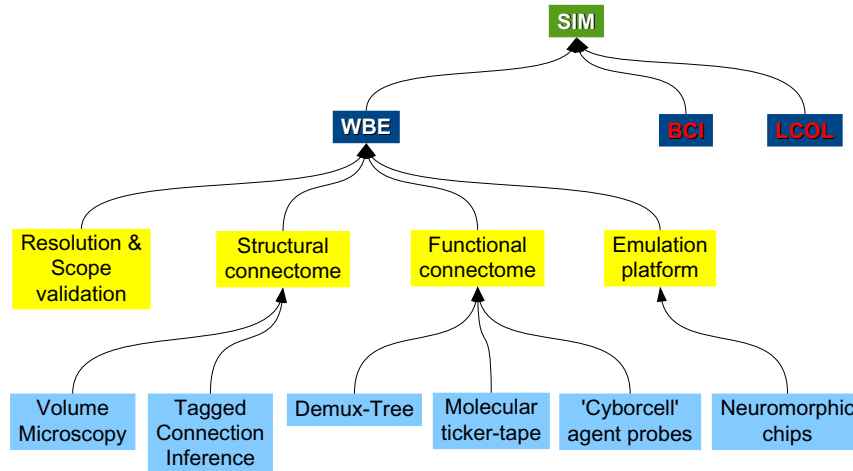


Fig. 4. A route to substrate-independent minds (SIM) via whole brain emulation (WBE). Four main requirements are distinguished: Investigations validating decisions about scope and resolution of data required for an emulation, structural connectomics that can identify latent function in the physical connectivity, functional connectomics which here includes both the functional characterization of components and the possible derivation of functional connections, and a sufficiently powerful computing platform for an emulation. Several current and ongoing project directions that may satisfy the requirements are indicated: Volume Microscopy projects similar to or based on the automatic tape-collecting lathe ultramicrotome (ATLUM), tagged connection inference by using DNA barcodes delivered to pre- and post-synaptic cells at each synapse and similar approaches, demultiplexing (Demux) tree concepts using branching nanowires or variants of the concept, *in-vivo* recording of events on implementations of molecular “ticker-tape”, *in-vivo* measurement by agent probes at cellular scale (Cyborcells) using infrared powered integrated circuits, and the development of neuromorphic hardware. At a minimum, there should be one successful project in each category, though some of the projects may also be applicable in conjunction. Brain–computer interfaces (BCI) and loosely-coupled off-loading (LCOL) are indicated in this graph, but not detailed as alternative routes that contribute to achieving SIM.

representation. There we find the application of technologies such as virtual reality, systems of machine learning that attempt to acquire your behavioral data at the sensory-motor or body and brain levels, data collection by introspective methods such as in the Bainbridge–Rothblatt approach, and the personalization of parameters of general cognitive architectures at the level of brain modules and cognitive functions.

Brain–Computer interfaces takes into consideration opportunities at a broad range of representation levels, from augmented reality devices, EEG/fMRI and TMS to neural probes based on individual or many electrodes and optogenetic stimulation. A successful WBE should certainly achieve reimplementations and enable interfacing, but the technological focus is presently on the reimplementations side. BCI starts on the interfacing side, but as the performance requirements increase that leads to many of the same problems faced in WBE. Due to the technologies considered, LCOL also begins as a form of interface, yet it is clearly intended to lead to reimplementations.

New tools with which to obtain the *structural connectome* include leading efforts by Ken Hayworth and Jeff Lichtman (Harvard University) building so-called Tape-to-SEM devices, such as the ATLUM [Hayworth, 2012] and its redesign to use FIBSEM [Knott *et al.*, 2008]. The ATLUM is a solution to the problem of collecting all of the ultra-thin slices from the volume of a whole brain for imaging by electron microscopy (EM) (Fig. 4, Volume Microscopy). Winfried Denk (Max-Planck) and Sebastian Seung (MIT) popularized the search for the human connectome [Seung, 2012]. The Denk group, using SBFSEM [Denk and Horstmann, 2004], has contributed to milestones such as the reconstructions by Briggman *et al.* [2011]. In the area of automated optical microscopy, projects are seeking to combine multiple channels of observation into a single comprehensive data stream [FARSIGHT Project, 2011]. The effort to address the automated collection of structure data from the whole brain was led by the laboratory of Bruce McCormick (now headed by Yoonsuck Choe, Texas A&M). The resulting knife-edge scanning microscope (KESM) can image the volume of a brain in a reasonable amount of time, but it cannot directly see individual synapses [McCormick and Mayerich, 2004]. Another newcomer with promise is the emerging field of X-ray microscopy [Yamamoto and Shinohara, 2002].

The ATLUM is the first stage in a series of designs for Tape-to-SEM that are meant to handle the volumes of neural tissue required for WBE. By comparison, the SBFSEM methods employed by Winfried Denk have sufficient resolution, but cannot handle volumes with which one could extract local circuitry and connections with neuronal circuitry in other brain modules. Another strength of the ATLUM is that it produces a library of original brain slices without loss of information from cutting that would affect a 3D reconstruction. The library affords random access for imaging.

Sectioning, imaging and reconstructing an entire brain using the ATLUM takes a very long time, though improvements are possible by combining the automated sectioning with new developments in automated high-throughput electronic microscopy. Furthermore, sectioning requires careful preparation of a brain in such a manner that all of the critical ultrastructure is conserved. Perfecting the methods of preparation is the theme of the Brain Preservation Technology Prize [Hayworth, 2011], the candidates for which are to be evaluated at our EM laboratories at Halcyon Molecular in 2012. Mapping 3D reconstructions to functional representations requires either a detailed structure-to-function mapping library or that the structural scan is complemented with functional characterization carried out at high resolution and large scale. A mapping library would have to contain functional components, their morphological features and mapping functions that deduce parameter values from variations in the morphology. Initial and successful efforts along these lines were made in Briggman *et al.* [2011]. Ways to obtain complementary functional data are described below.

Groups led by Anthony Zador (Cold Spring Harbor Laboratory) and Ed Callaway (Salk Institute) have chosen an entirely different route to infer high resolution full connectome data (Fig. 4, Tagged Connection Inference). As mentioned earlier, Zador

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proposes a number of ways to deliver unique DNA “barcodes” to the pre- and post-synaptic neurons of each synaptic connection [Zador, 2011; Oyibo *et al.*, 2011]. Neuronal cell bodies are extracted and DNA is recovered from each. In a promising variant of the protocol, pre- and post-synaptic neuron-specific RNA barcodes connect to neurexin and neuroligin at each synapse. A biotin label attaches to neurexin and neuroligin at each side of the synapse [Thyagarajan and Ting, 2010] and that is used to pull out the complex with both RNA codes. By identifying the specific sequences within, it should be possible to find matching pairs that act as pointers between connected neurons. Note the level of representation that matches this tool. The order of synapses on a dendrite and other details about the synapses are not retrieved with the first versions of this method.

To obtain the *functional connectome*, we want to measure and record changes of relevant signals at each neuron in the brain. The number of targets is very large, but they form a network. A general interfacing concept that takes this into account is the so-called Demux-Tree or demultiplexing tree (Fig. 4, Demux-Tree). The earliest implementation proposals for such trees involve the use of nanowires inserted through the vasculature that sprout branches reaching every neuron [Watanabe *et al.*, 2009]. The branching of wires remains to be accomplished, though the flexible nanowires have been developed at the New York University School of Medicine with a width of only 500nm. Even so, the total quantity of wire would displace a considerable volume within the brain. Alternative Demux-Tree concepts would build only the end-nodes and possibly branching vertices of the tree, not the branch edges. In a node-only tree, the nodes would communicate wirelessly or by transmitting through neuronal arbor signals of a kind that neurons are relatively insensitive to (e.g., electric signals at very high frequency, ultrasound, etc.).

A collaboration between labs at MIT (Ed Boyden), Northwestern University (Konrad Kording), Harvard University (George Church), and Halcyon Molecular (Randal A. Koene and Rebecca Weisinger) is exploring an approach at even smaller scales by taking advantage of biological means of signal processing. This approach seeks to record functional events at the molecular level, resulting in a kind of “molecular ticker-tape” (Fig. 4, Molecular ticker-tape).

Working with biological machinery is still difficult, often involving a search for useful components such as voltage dependent receptors and ways to influence their operation without adverse effects to the cell. Non-biological nanotechnology has not advanced to the point where we can reliably construct devices. What we do understand very well is how to work with integrated circuits. Therefore, combining the ideas of hierarchical measurements in a node-only Demux-Tree with the desire to probe from within at the cellular scale, new efforts include the development of micron-scale devices based on existing integrated circuit fabrication to be used *in-vivo* in large quantities (Fig. 4, Cyborcell agent probes). Using integrated circuits at the cellular level has been tested by pioneers such as Gómez-Martínez *et al.* [2010], and an implementation concept that includes RFID-like passive communication with

the outside is being worked on by Yael Maguire and George Church (Harvard University). These circuits may be powered and receive instructions *in-vivo* by transmissions at infrared wavelengths. We describe the molecular ticker-tape and integrated circuit projects in greater detail below.

Before those tools to obtain a functional connectome reach maturity, it is possible to carry out high resolution functional characterization with electrode arrays. Current methods in neuroscience rely heavily on single unit recording with fluid filled glass electrodes or on recording with planar or penetrating multi-electrode arrays (e.g., the 4096 CMOS electrode 3Brain/Plexon BioCAM system). These methods are popular, because the signals recorded provide significant dynamic information. Buzsáki [2004] presented a method of recording with many silicon electrodes to capture the activity of all neurons within a 200 μm cylinder, while stimulating optically. They speculate about placing many such recording cylinders at 200 μm distances to eventually achieve full brain coverage. There are ongoing efforts in the Boyden lab to use micro-electrode arrays with thousands of recording channels that incorporate light-guides for optogenetic stimulation [Henninger *et al.*, 2011; Zorzos *et al.*, 2011a,b; Kondandaramaiah *et al.*, 2011]. A stimulation-recording array of that kind can perform sensitivity analysis *in-vivo* and explore hypotheses of great relevance to WBE. Peter Passaro (U. Sussex) is working on an automation scheme for research and data acquisition aimed at WBE. Suitable modeling conventions are inspired by neuro-engineering work of Chris Eliasmith [Eliasmith, 2008; Eliasmith and Anderson, 2003]. Meanwhile, Ted Berger (USC) is continuing his work on cognitive neuroprosthetics [Berger *et al.*, 2005, 2011], which forces investigators to confront challenges in functional interfacing that are also highly relevant to WBE.

3. Solving Whole Brain Functional Characterization *In-Vivo*

Let us consider the number of components involved in a WBE strategy of straightforward duplication of mental activity. The human brain has up to one hundred billion (10^{11}) neurons and between one hundred trillion (10^{14}) and one quadrillion (10^{15}) synapses. Note though, that we have reached a point in technology development where, for purposes of data acquisition, these objects are now considered fairly large (e.g., 200–2,000 nm for synaptic spines and 4,000–100,000 nm for the neural soma). At least, this is true by the standards of the current nanotechnology industry that works with precision at 10–100 s of nanometers. In terms of their activity, the components are mostly quiet. The temporal and spatial scope and resolution that are required for data acquisition are not easily achieved with external imaging modes. The power or field strength (e.g., in the case of high field fMRI) of external imaging modes must increase to harmful levels to achieve the resolution, especially when measurements need to be taken over extended time intervals.

Interfacing and recording at large scale and high resolution may be better accomplished from within the brain. The methods available then operate in direct analogy with the biological system of information exchange that is already in place.

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The substrate of the brain itself demonstrates the approach, and similar modes of signal detection, measurement, recording and delivery can be applied. Within the brain's substrate, many specialized cells already collaborate as "agents" to accomplish brain mechanisms that work with the information we are interested in. Other operators include members of the human microbiome (bacteria, fungi, archaea), many of which perform useful tasks for the human host, as well as invasive microorganisms (bacteria, protists) and virii that reach and interact with sites of interest. DNA programming, engineered cells or synthetic bacteria are an active topic of research in laboratories such as those of George Church [Tian *et al.*, 2004; Forster and Church, 2006a,b] and the J. Craig Venter institute [Gibson *et al.*, 2008]. It may even be possible to reutilize features of neuronal physiology itself to provide interface and observation channels (e.g., compare with work by Choi *et al.* [2006] to create pace-makers by growing skeletal muscle cells). We see early work in this direction in the design of neurotrophic electrodes [Kennedy *et al.*, 2000].

Ultimately, we would like to be able to construct specialized nanoscopic agents, using methods of self-organization or fabrication by conventional mass-production. Similar approaches have been used in the production and application of nanowires intended for interfacing with neuronal circuits [Cohen-Karni *et al.*, 2009], but engineered systems or synthetic biology at molecular and meso-scales are areas still in need of fundamental research and development. We can start with heftier components, utilizing manufacturing methods where great expertise exists today, as described in following sections. This comparatively primitive technology will still suffice to produce tools far better suited to *in-vivo* high resolution measurements in the whole brain than any of the tools that have so far been employed in neuroscience.

3.1. *Biological event-recording, a.k.a. the "molecular ticker-tape"*

The basic idea is to use a biological means of registration of interesting events, such as changes in membrane potential. The rationale is that such means already exist at the resolution required and can be delivered at the scale of a whole brain. The biological storage medium is here called a "molecular ticker-tape". One such medium is DNA, for example as described in the patent application by Church and Shendure [2010]. Using such storage, a method for massive long-term low-pass readout of neuronal activity has been proposed by Konrad Kording (Northwestern University), by encoding changes in calcium molarity. Similarly, discussions between Ed Boyden (MIT) and researchers at Halcyon Molecular led to a proposal involving the storage of the high frequency dynamics of neuronal responses, by using voltage-gated channels or voltage-sensitive dyes and proteins to trigger encoding. Correspondingly, a collaboration has emerged between the Church, Boyden, Kording and Halcyon Molecular labs to explore and develop molecular ticker-tape technology.

Molecular encoding, such as onto DNA, can make use of several mechanisms. We note three possible mechanisms here: (1) voltage-dependent error-encoding during rolling circle amplification of DNA, (2) voltage-gated oligo release and amplification

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(the “copy number” measure of time), and (3) voltage-gated cutting of released DNA (the “footprint size” measure of time). There are many other mechanisms that could be utilized, for example the synthesis of RNA or DNA polymers, the modification of DNA by methylase, oxidation by charge transport, glycosylase, deaminase, or the modification of peptides by kinase, phosphatase or amidoligase. Also, the signals used to affect the mechanism can include voltage, calcium and many others. Whatever the implementation of the molecular ticker tape, the mechanism of its encoding, or the triggering signals, the tapes need to be collected for read-out. In the case where DNA is deposited within the cell bodies of neurons, the neurons are extracted and all of the DNA snippets found therein identified, measured, and possibly sequenced.

The “copy number” measurement of time between events is shown in Fig. 5(a). Liposomes deliver oligos to the neurons. Voltage-gated oligo release takes place, and the oligos that collect within the neuron are amplified in-cell. The more copies of a specific oligo are found, the longer it has been there, which provides a measure of time since release. Figure 5(b) shows the “footprint size” measurement of time between events. Multiple phages bind to each neuron and release DNA rapidly and robustly in a photo-gated manner. DNase is activated in a voltage-gated manner, which causes cutting of the inserted DNA — but that cutting is incomplete with some probability distribution. Consequently, a distribution of cuts is obtained for different voltage events, and the resulting lengths of collected DNA indicate the times since onset of the experiment at which voltage events occurred. It is clear that the implementations described here are aimed at short periods of characterization at high temporal resolution.

In support of the feasibility of this technique, it is useful to note that DNA and RNA transcriptional switches have already been designed to implement bistable dynamic behavior and oscillation [Kim *et al.*, 2006]. And certain complex regulatory networks have already been engineered and incorporated in living cells. For example,

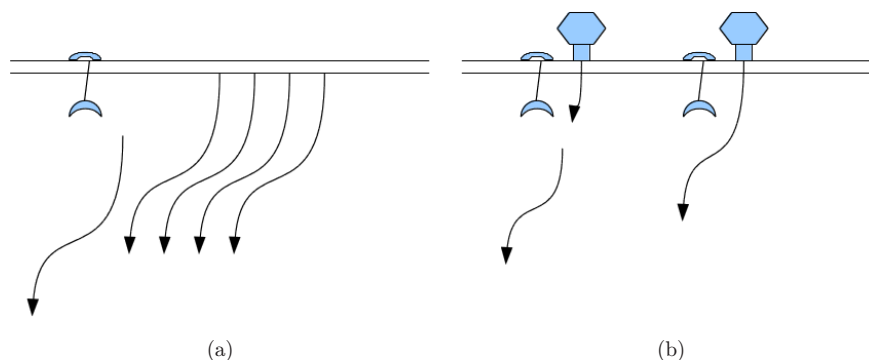


Fig. 5. Two mechanisms for event-triggered encoding on DNA “molecular ticker-tape”. (a) Measuring time between events in terms of “copy number”. (b) Measuring time between events in terms of “footprint size”. The double lines represent the cell wall. Curved arrows are strands of DNA. Hexagons are phages. The half-moon is a voltage-gated channel that cuts or causes release of the DNA strands.

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Elowitz and Leibler [2000] created a synthetic oscillator that periodically changes the color of a cell based on transcription regulators.

Some of the questions that the collaborative effort needs to answer are: What type of signals and events should be recorded? For example, we may wish to encode quantitative measures of membrane potential in addition to the timing of spiking events. What is the encoding rate that is needed in order to usefully record the timing of spikes in a neuron's spike train, or the changes of a neuron's membrane potential during a neural response? Is it possible to use event-triggers and variable encoding rates, or should encoding be clocked? What is an acceptable error rate for the encoding on one "tape"? How may differences of the electrical properties across a cell body affect the aggregation of meaningful cell-specific information?

The temporal alignment of tapes collected from multiple cells will require some synchronizing mark, such as an initial encoding triggered by a synchronizing event (e.g., an optical signal). Synchronization need not span the whole of the regions in which recording takes place, as long as there is a hierarchical strategy by which locally synchronized tapes can be aligned globally. Initiation of molecular recording and synchronization signals should not be harmful. And of course, it will be preferable if a way is found to harvest the molecular ticker-tapes *in-vivo* rather than *post-mortem*. Eventually, it will be very useful to close the loop between (possibly optogenetic) stimulation and molecular recording in a dynamic manner, so that the process constitutes a biological brain–computer interface.

3.2. *Electronic agent systems: artificial cells and a wireless computing cloud co-residing in the brain*

Earlier, we mentioned the ideal of nanoscopic agents or "nanobots" that could collect information from within the brain *in-vivo* and that could form bi-directional interfaces with the biophysical processes therein. Nanotechnology is not ready for that. The dimensions of such agents and their replication all over the brain are features offered by biological solutions. We are able to devise schemes such as the molecular ticker-tape, which will certainly continue to improve and to remove existing limitations of the technique. At present, our expertise at synthetic biology and systems biotechnology does not yet suffice to carefully manipulate biology so that we can readily produce hierarchical strategies for the extended collection of data *in-vivo* without adverse effects. We cannot yet construct and program arbitrary biological solutions within safety constraints.

What we do have is a lot of experience working with standard integrated circuit (IC) technology as developed in the semiconductor industry. We can build devices to spec and we can arrange networks and hierarchies of cooperating components to accomplish complex strategies. IC technology is still shrinking (Fig. 6): 32 nm technology appeared in the mid 2000's and entered commercial use in early 2010; 22 nm technology was developed in 2008–2009 and Intel produced a microprocessor based on the technology in 2011. Between 2013 and 2015, further shrinkage to 16 and

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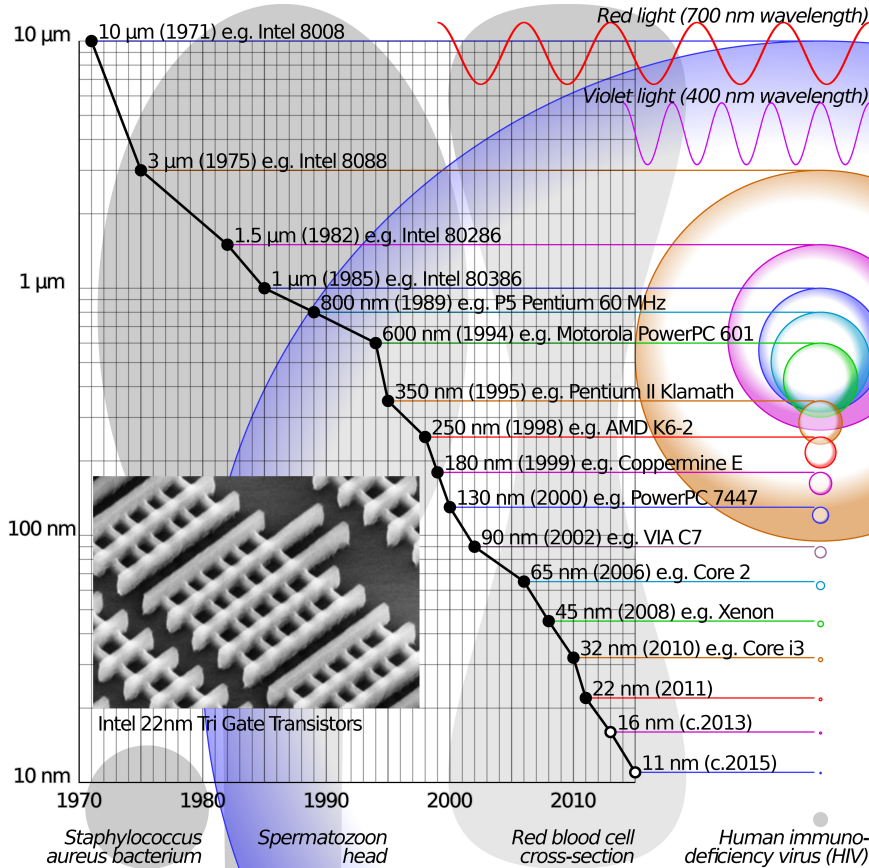


Fig. 6. Past, current and predicted semiconductor process node sizes. Gray background shapes illustrate comparative sizes in biology. The electron microscope image shows multiple Tri Gate transistors produced with Intel's 22 nm technology. (Graph by Wikipedia author Cmglee, Creative Commons Attribution-Share Alike 3.0 Unported license. Electronmicroscope image of Tri Gate transistors copyright Intel Corporation, with permission.)

11 nm/8 nm process nodes are anticipated. IC's are being patterned in many layers, adding a virtual third dimension and thereby further increasing the density of computation. There are other technologies poised to replace this silicon IC technology in the next few years [Avouris and Chen, 2007; Changxin and Yafei, 2009], but even with the current (soon to seem prehistoric) technology we can address the fundamental needs of WBE.

At the 32 nm process node, an $8 \mu\text{m} \times 8 \mu\text{m}$ die can contain 2100 transistors, which is approximately equal to the number of transistors on Intel's first general purpose CPU, the i4004. The i4004 is Turing complete. Using Intel's new 3D Tri-Gate transistors in 22 nm process technology, the count nearly doubles to just under 4200 transistors. Anticipated 16, 11 and 8 nm process nodes may put about 7900 (more than in the i8080 used in cruise missiles), 16.7 K and 31.6 K transistors

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(more than the i8088 used in original IBM PCs) on the die respectively. The $8\ \mu\text{m} \times 8\ \mu\text{m}$ die size is chosen here, because it compares directly with the $8\ \mu\text{m}$ diameter of red blood cells, which make their way to every neuron in the brain through the capillaries of the vasculature. These dimensions are still clunky by the standards of molecular processes or true nanotechnology, but they are dimensions at which operating *in-vivo* in large numbers throughout a brain is feasible. The computing capacity is sufficient to carry out measurements, calculations, temporary storage and communication.

Integrated circuits at cellular scale have been developed and even inserted into living cells without causing any disruption of cellular processes. Gómez-Martínez *et al.* [2010] successfully internalized silicon chips within *D. discoideum* amoeba cells and human HeLa cells. The chips had a lateral dimension of $1.5\text{--}3\ \mu\text{m}$ and a thickness of $0.5\ \mu\text{m}$. *D. discoideum* cells are phagocytotic, having the ability to engulf particles outside the cell, so they easily inserted the chips. Human HeLa cells however are not phagocytotic, so Gómez-Martínez *et al.* used a lipofection technique to transfer the chips inside the cells. Once embedded by phagocytosis, lipofection or microinjection, the microchips did not disrupt cells' viability and there was no apparent toxicity. The researchers showed that the integrated circuits could be used as intracellular sensors. The same team is also developing means for attaching and keeping in place microdevices within organic environments through a procedure they call nanovelcro [Durán *et al.*, 2010]. Compared with micro- and nanoparticles used in imaging, intracellular silicon chips have many potential advantages: Nanometric precision in shape and dimensions, integration of many different materials with different dimensions and geometries, 3D nanostructuring, integration of electronics, integration of mechanical parts, all the advantages of microelectromechanical systems (MEMS) and nanoelectromechanical systems (NEMS).

Eventually, the glucose used for energy within the body may also be used to power integrated circuits, as demonstrated through an implanted glucose biofuel cell in Cinquin *et al.* [2010]. While promising, that is not presently a mature technology, ready for application at the micron scale. Initially, we may supply power through pulsed infrared radiation, as there is a window of organic transparency for wavelengths between 200 and 900 nm [Hale and Querry, 1973; Cooper *et al.*, 1994; Hollis, 2002], though the optimal wavelengths for gray and white matter in the brain need to be established. The initial goal is to deliver a very large number of these IC devices through the vasculature of a brain.

Extreme flexibility is needed for positioning of measurement agents, so that the solutions with the greatest potential all involve wireless communication. Low-power methods of bi-directional communication are inspired by the success of passive radio-frequency identification (RFID) technology. Yael Maguire [Greaves *et al.*, 2007], one of the main contributors to that field, is now involved in the efforts of the Church lab (Harvard University) to create prototype ICs for *in-vivo* neural recording. In this case, radio frequencies are not well suited to the task, so that the RFID-like construct will instead be designed to work with infrared frequencies. Maguire has called this a micro-OPID. In Fig. 7, we see the conceptual layout of the infrared powered IC.

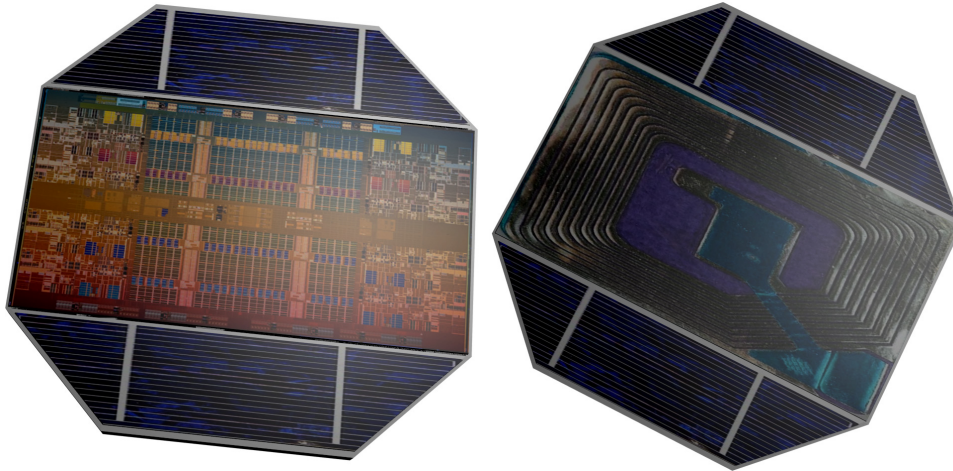
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Fig. 7. Front and back sides of the $8\ \mu\text{m} \times 8\ \mu\text{m}$ agent device based on IC technology. Semiconductor layout is shown along the central strip of the die. Infrared power collection and IR communication reception may be implemented on the wings front and back. The central portion of the backside of the die is reserved for a passive communication antenna intended to operate at IR wavelengths.

Packaging of the ICs for *in-vivo* use can be bio-compatible or bio-active. Bio-compatible packaging can consist of protective encasing in silicon or other safe inorganic materials, or it can involve embedding within cells or protein shells such as recently constructed artificial red blood cells [Doshi *et al.*, 2009]. The surface of the envelop can be functionalized to be bio-active or bio-mimetic in useful ways. Within its biocompatible and possibly functionalized packaging, we call these agents Cyborcells (Fig. 8) in reference to their dual nature, similar to the bio-machine duality of the cyborg concept, but at the scale of cells.

Even at $8\ \mu\text{m}$, the size of this first target may be impractical for many of the requirements of *in-vivo* high-resolution large-scale interfacing and acquisition. Red

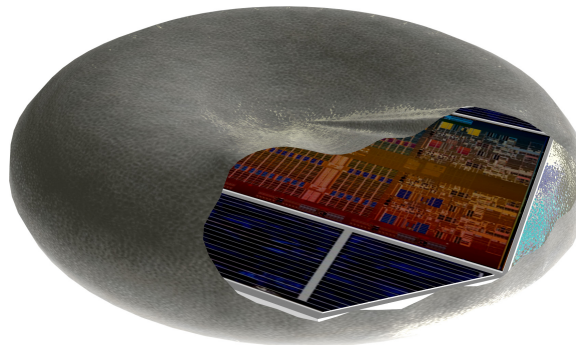


Fig. 8. A "Cyborcell" composed of the IC agent device within a bio-inert or bio-active casing. The Cyborcell shown here has the general dimensions and shape of a red blood cell.

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blood cells make their way through the smallest capillaries by deforming their torus shape. Deformable containers for *in-vivo* use have been developed. In fact, the artificial red blood cells created by Doshi *et al.* [2009] mimic the key structural and functional features of red blood cells. They were shown to be able to carry oxygen, flow through capillaries smaller than their own diameter and encapsulate drugs and imaging agents. ICs do not deform in that manner. In any case, we want to access smaller spaces, navigating the cerebrospinal fluid and interstitial fluids directly, passing between glial cells, possibly following axonal channels [Raper and Mason, 2010] and taking advantage of other means of transportation, such as guidance by externally applied magnetic fields. For this reason, the project calls for a more flexible and cooperative strategy between multiple agent types. This closely approximates a data acquisition concept for SIM that has been called “internal agent collaborative scanning with external imaging” (IACSEI) in previous accounts [Koene, 2011].

A team of agents, as in Fig. 9, may consists of larger hub cells that deal with complex computational tasks such as logistics, determining the relative locations between agents in a local network, collecting and temporarily storing data, communication and cloud computing in coordination with other hubs. Some hubs may be involved in harvesting energy *in-situ* from glucose [Cinquin *et al.*, 2010] to operate as local transmission sources to power communication and smaller agents. Smaller agents act as sensor cells, stimulation cells, or have tasks that involve herding other agents, forming physical chains, the delivery of biological tags (e.g., DNA “barcodes”), and detecting local morphological features that contribute to a structural connectome. Physical interaction with the surrounding physiology may be achieved through micro-actuators, functionalization of the casing of a Cyborcell, and through the use of nanoscopic materials such as nano-velcro [Durán *et al.*, 2010].

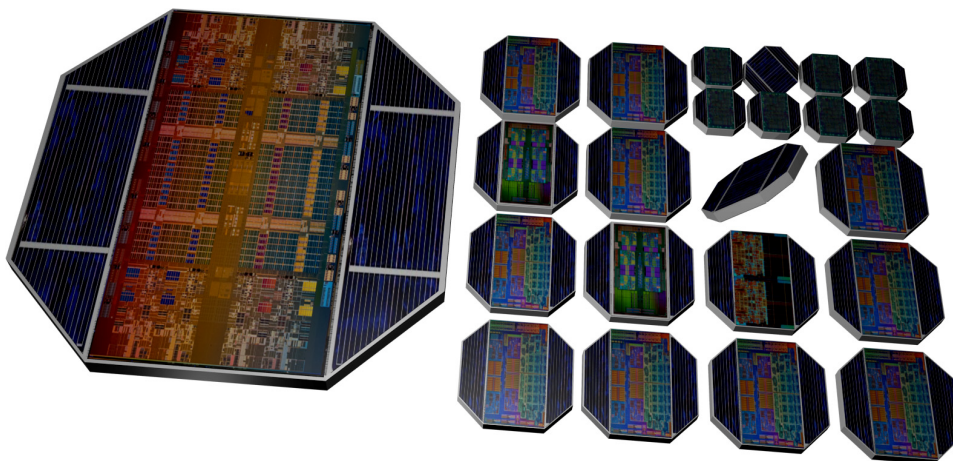


Fig. 9. A depiction of an example of a collaborating team of agents. The hub agent IC shown has $8\ \mu\text{m} \times 8\ \mu\text{m}$ dimensions, the smaller agents have dimensions of $2\ \mu\text{m} \times 2\ \mu\text{m}$ and $1\ \mu\text{m} \times 1\ \mu\text{m}$. In the team shown, only the large hub has a passive communication antenna.

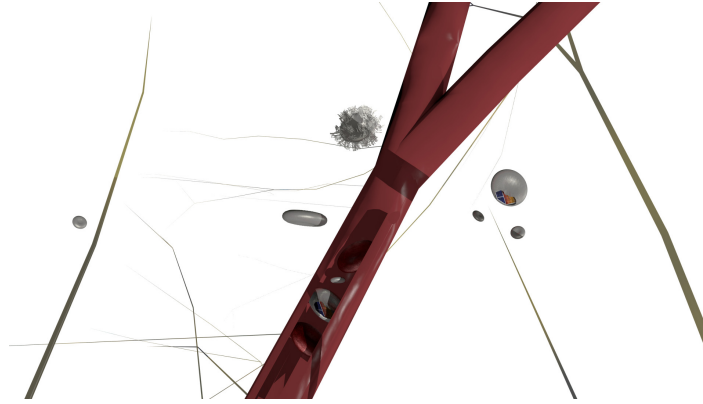
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Fig. 10. Eight Cyborcells (gray) collaborating within the environment of neuronal circuitry. One hub and one smaller agent are shown within a capillary of the vasculature, surrounded by red blood cells. Three other hubs and three smaller agents are shown within the interstitial spaces around dendritic arbor (grayish fiber branches). One of the hubs is functionalized with nano-velcro. The depiction is sparse, as it does not show volumes occupied by other cells such as glia.

Several hubs and smaller agents are depicted within vasculature and in the cerebrospinal fluid around dendritic arbor in Fig. 10.

Neuronal currents, conductance or voltage changes may be detected with or without contact. The capacitance across on-chip transistors may be used to detect changes of the electric field. We may detect the flow of ions, or pick up the electric field changes across the membrane in the manner of a voltage sensitive dye. Cyborcells may also operate in conjunction with perfused voltage sensitive dye, locally observing the spectral shift of the dye. Differences may be triangulated between multiple agents, and conduction through cerebrospinal fluid, interstitial fluid or on the surface of cell bodies may be used to ground a common reference. A micromechanical patch-clamp may be feasible when necessary, and when a better reference electrode is required then a few special, larger hubs may be designed to operate as shared electrodes. At the dimensions depicted, designating one small agent to each of 10^{11} neurons and one hub to every 10 small agents requires a volume of 1.04 cm^3 . As shown in Fig. 11, that is $1/1635$ th of the volume of a typical human brain.

The Cyborcells form what is in essence a hierarchical computational network that resides within and operates concurrently with the living and functioning brain network. The artificial network can store and process information and becomes an access interface to characteristic component functions at the scale and resolution needed for WBE. While cruder than envisioned nanoscopic technology [Sierra *et al.*, 2005], the methods have similar potential and the core integrated circuit technology is familiar and exists now [Bohr and Mistry, 2011].

It is generally preferable to implement a measurement network that causes no unintended interference with the operation of the existing biological network. For this reason, the hierarchy of recording devices is conceived primarily as a system of

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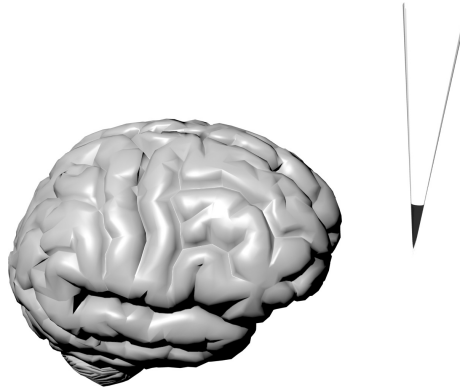


Fig. 11. The comparative volumes of the human brain and a vial containing over one hundred billion Cyborcells (silver-gray volume at the tip of the vial). Diagnostics may be carried out between Cyborcells prior to insertion. For example, network coordination may be tested by running a 700 city traveling salesman problem on a cloud of one million Cyborcells that together consume a volume of well under 0.008 mm^3 .

nodes without physical edges (node-only Demux-Tree) and utilizing a different medium, such as pulsed infrared or ultrasonic means of communication between the devices. The node-only approach can be easier to set up as well. It is nonetheless worthwhile to consider those cases where physical communication chains are advantageous. For example, stringing a wire between nodes, or even utilizing the electrical conductivity of cerebrospinal fluid may provide a better means of establishing a common ground for precise voltage measurements. In addition to inferred functional connectivity and possible morphological measurements from within, we may apply a volume microscopy technique such as Tape-to-SEM to find the precise locations of Cyborcells within brain tissue. Of course, that approach is not *in-vivo*.

Once we have a controllable hierarchical network of agents operating concurrently with the brain then we can design their actions to suit our requirements for data acquisition. We can address sensing and stimulating modes beyond those included in the initial concept. For example, another type of signal that has been measured on-chip is oxygen level. This is done using Clark-Type electrodes, which have already been engineered down to the $10 \mu\text{m}$ scale. We need to design the ways in which recording agents detect and measure those signals that are required.

3.3. Delivery and function

Whether of biological or electronic origin, agents may be delivered to sites within the brain following injection or infusion into the cerebral vasculature, cerebrospinal fluid and interstitial fluids. The aim will be to achieve recording at very many sites, and delivery of the agents may be targeted or follow a simple distribution scheme.

Responses that are generated by collaborating agents may be detected directly through imaging and stored externally, or the responses may be handed to delivery

agents. Delivery agents may be recovered by cycling out through the vasculature, or they may form communication chains.

A very interesting further aim is to create a class of agents that act as stimulation probes, able to replicate recorded responses in-place. Sets of such stimulating agents could provide neuroprosthetic function.

With a large-scale distribution and location registration, morphological structure may be inferred from recorded data, via stimulation comparable to that used in circuit analysis [Sakai, 1992]. *Post-mortem* acquisition of a structural connectome can still be done following an intravascular procedure.

4. Conclusions

Whole brain emulation does not require us to have a full understanding of the brain, where such an understanding would include knowledge of the strategies employed at each level in a hierarchy of functions of the mind. Instead, we need to be able to characterize the functional behavior of fundamental computational components (e.g., neuronal responses, post-synaptic responses, diffuse modulation), and we need to be able to acquire component data and connectivity between components. This is possible with today's knowledge and with tools built using today's technology. While not all neurons, synapses or channel types have been fully characterized and cataloged, doing so is a regular activity in neurophysiology and we know how to accomplish it where and when it is needed.

Whole brain emulation efforts face a couple of core problems. The first such problem is to test our assumptions about contributing factors at scope and resolution. What is the required information about functions of the mind that is contained in their biological implementation? For example, do we need to identify methylation and demethylation states of DNA in neurons [Day and Sweatt, 2010] or is the effect of these processes adequately covered by identifying other synaptic changes such as their morphology or their conductance response? This was specified in terms of the need to identify sets of elements and relations that are expressed in $\mathbf{x}(t)$, $\mathbf{f}(t, x(t), u(t))$ and $\mathbf{g}(t, f(t), x(t), u(t))$, and in terms of the need to quantify and measure their parameter values. We should also consider if the answers to these questions generalize to all brains, or which types of subject specific differences there may be. It is likely that the answers are not the same for all modular parts or functions of the brain, so that we are faced with the need to integrate multiple scales.

We can constrain the problems of scope and resolution by looking at physical limitations that play a role in the data that is processed by some sub-system of the brain. Those can be physical limitations of the biophysics of the machinery that is carrying out the processing, and they can be limitations of the data received from the environment or the use of data transmitted to the environment. The limits depend on emulation choices, as represented by the target time interval and precision of computation. In effect, we constrain the bottom-up reconstruction in accordance with a

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(partial) top–down understanding of system strategy and operational requirements. The problem of data acquisition can be reduced further by recognizing ways in which a smaller number of measurements can be used to infer category-specific characteristic features of the computational components in an emulation.

The second core problem is the question what sort of technology is needed, available or should be developed in order to make it possible to acquire the requisite data. At present this appear to be a race between advances in materials science vs. engineering in cell biology. Eventually, the answer may be a combination of both, providing collaborating biological and electronic intra-vascular agents, which make data at large scale and high resolution visible to aggregation methods based on developments in imaging technology. An aggregation phase, as well as wireless or passive transmission of measurements are common elements of the most promising new measurement tools. Emission and detection can be optimized to each other and do not depend on the ability to detect externally the biophysical variables to be measured.

The obvious next step, after creating agents that can reside within the brain in large numbers and that form a secondary computing network within the brain, is to have agents that can stimulate neurons, thereby closing the loop. At that point, you have a very large scale brain–computer interface. That interface would not only enable augmentation, it could provide neuroprosthetic function by replicating recorded responses in-place. The development of such an interfacing method can benefit from many of the same sources of market-pull as the overall development of brain–computer interfaces. Even crude, initial versions can be used to achieve valuable results. Each iteration of the development cycle offers new opportunities in terms of spin-out technology with a higher bandwidth and more intimate connection between the human brain and diagnostic, augmentative and actuating devices.

Reimplementation can in principle take place in any platform that can carry out the requisite computations, but some platforms will be more well-suited to the task. For example, we imagine that a neuromorphic hardware may be more readily adapted to carry out the emulated functions of a whole brain.^a

The most important message to take from this paper is that a program to achieve WBE is well-conceived and feasible in the foreseeable future by a combination of concrete projects that use existing technology or are presently being developed based on our current scientific and technological understanding.

^aNote that the suitability of a processing substrate has very little to do with perceived separation of hardware and software. It is often said that neural networks and the neuronal network of the brain differ from traditional Von Neumann computers in that there is no clear distinction between hardware and software. Consider though, that the distinction is somewhat arbitrary in traditional computers as well, since we may choose to create a function in program software, or we may embed it in the hardware such as in a field-programmable gate array (FPGA). More important distinctions are the vast numbers of concurrent processors in a brain and their event-triggered processing.

References

- Aho, A., Donner, K. and Reuter, T. [1993] "Retinal origins of the temperature effect on absolute visual sensitivity in frogs," *Journal of Physiology* **463**, 501–521.
- Atanasijevic, T. *et al.* [2006] "Calcium-sensitive mri contrast agents based on superparamagnetic iron oxide nanoparticles and calmodulin," *Proceedings of the National Academy of Sciences* **103**(40).
- Avouris, P. and Chen, Z. [2007] "Carbon-based electronics," *Nature Nanotechnology* **2**(10), 605–615.
- Bainbridge, W. [2006] "Strategies for personality transfer," *Journal of Personal Cyberconsciousness* **1**(4), http://www.terasemjournals.org/PCJournal/PC0104/bainbridge_01a.html.
- Bartels, J. *et al.* [2008] "Neurotrophic electrode: Method of assembly and implantation into human motor speech cortex," *Journal of Neuroscience Methods* **174**, 168–176.
- Berger, T. *et al.* [2005] "Restoring lost cognitive function," *IEEE Engineering in Medicine and Biology* **24**(5), 30–44.
- Berger, T. *et al.* [2011] "A cortical neural prosthesis for resoring and enhancing memory," *Journal of Neural Engineering* **8**(4), doi: 10.1088/1741–2560/8/4/046017.
- Bialek, W. *et al.* [1991] "Reading a neural code," *Science* **252**, 1854–1857.
- Bock, D. *et al.* [2011] "Network anatomy and *in vivo* physiology of visual cortical neurons," *Nature* **471**, 177–182.
- Bohr, M. and Mistry, K. [2011] "Intel's revolutionary 22 nm transistor technology," Tech. rep., Intel.
- Boyden, E. *et al.* [2005] "Millisecond-timescale, genetically targeted optical control of neural activity," *Nature Neuroscience* **8**(9), 1263–1268.
- Briggman, K., Helmstaedter, M. and Denk, W. [2011] "Wiring specificity in the direction-selectivity circuit of the retina," *Nature* **471**, 183–188.
- Buzsáki, G. [2004] "Large-scale recording of neuronal ensembles," *Nature Neuroscience* **7**(5), 446–451, doi: 10.1038/nrn1233.
- Changxin, C. and Yafei, Z. [2009] "Nanowelded carbon nanotubes: From field-effect transistor to solar microcells," in *Nano Science and Technology Series* (Springer), p. 63.
- Choi, Y. *et al.* [2006] "Cardiac conduction through engineered tissue," *The American Journal of Pathology* **169**, 72–85.
- Church, G. and Shendure, J. [2010] "Nucleic acid memory device," Tech. Rep. US Pat. App. 20100099080, Harvard.
- Cinquin, P. *et al.* [2010] "A glucose BioFuel cell implanted in rats," *PLoS ONE* **5**(5), e10476, <http://dx.doi.org/10.1371/journal.pone.0010476>.
- Cohen-Karni, T. *et al.* [2009] "Flexible electrical recording from cells using nanowire transistor arrays," *Proceedings of the National Academy of Sciences USA* **106**, 7309–7313.
- Cooper, C. *et al.* [1994] "Near-infrared spectroscopy of the brain: relevance to cytochrome oxidase bioenergetics," *Biochemical Society Transactions* **22**(4), 974–980.
- Day, J. and Sweatt, J. [2010] "DNA methylation and memory formation," *Nature Neuroscience* **13**, 1319–1323.
- Deisseroth, K. [2011] "Optogenetics," *Nature Methods* **8**(1), 26–29.
- Denk, W. and Horstmann, H. [2004] "Serial block-face scanning electron microscopy to reconstruct three-dimensional tissue nanostructure," *PLoS Biology* **2**(11), e329.
- Doshi, N. *et al.* [2009] "Red blood cell-mimicking synthetic biomaterial particles," *Proceedings of the National Academy of Sciences of the USA* **106**(51), 21495–21499.
- Durán, S. *et al.* [2010] "Silicon nanowire to attach inorganic microdevices to biological material," in *Proc. 14th International Conference on Miniaturized Systems for Chemistry and Life Sciences*, Groningen, The Netherlands, pp. 503–505.

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- 1 Eliasmith, C. [2008] “How we ought to describe computation in the brain,” *Computation in*
- 2 *Cognitive Science Workshop*.
- 3 Eliasmith, C. and Anderson, C. [2003] *Neural Engineering: Computation, Representation and*
- 4 *Dynamics in Neurobiological Systems* (MIT Press, Cambridge, MA).
- 5 Elowitz, M. and Leibler, S. [2000] “A synthetic oscillatory network of transcriptional reg-
- 6 ulators,” *Nature* **403**(6767), 335–338.
- 7 FARSIGHT Project [2011] “Developing quantitative tools for studying complex and dynamic
- 8 biological microenvironments from 4d/5d microscopy data,” Tech. rep., <http://www.farsight-toolkit.org/>.
- 9 Forster, A. and Church, G. [2006a] “Synthetic biology projects *in vitro*,” *Genome Research* **17**.
- 10 Forster, A. and Church, G. [2006b] “Towards synthesis of a minimal cell,” *Molecular Systems*
- 11 *Biology* **2**, 45.
- 12 Gibson, D. *et al.* [2008] “Complete chemical synthesis, assembly, and cloning of a mycoplasma
- 13 genitalium genome,” *Science* **319**(5867), 1215–1220, doi: 10.1126/science.1151721.
- 14 Gómez-Martínez, R. *et al.* [2010] “Intracellular silicon chips in living cells,” *Small* **6**(4),
- 15 499–502.
- 16 Greaves, J., Sarma, S. and Maguire, Y. [2007] “The future of RFID,” Tech. rep., Technology
- 17 Review Webcast, Cambridge, MA.
- 18 Hale, G. and Querry, M. [1973] “Optical constants of water in the 200 nm to 200 μ m wave-
- 19 length region,” *Applied Optics* **12**, 555–563.
- 20 Hayworth, K. [2011] “Proposal for a brain preservation technology prize,” Tech. rep., The
- 21 Brain Preservation Foundation, [http://www.brainpreservation.org/docu-](http://www.brainpreservation.org/documents/proposalforbrainpreservationtechnologyprize.pdf)
- 22 [ments/proposalforbrainpreservationtechnologyprize.pdf](http://www.brainpreservation.org/documents/proposalforbrainpreservationtechnologyprize.pdf).
- 23 Hayworth, K. [2012] “Electron imaging technology for whole brain neural circuit mapping,”
- 24 *International Journal of Machine Consciousness* **4**.
- 25 Henninger, R. *et al.* [2011] “A novel concept for an implantable probe for deep-brain optical
- 26 measurement of the activity of large populations of neurons,” in *2011 Neuroscience Meeting*
- 27 *Planner* (Society for Neuroscience, Washington, DC, 2011). [Online.], p. Program No.
- 28 305.08.
- 29 Hollis, V. [2002] *Non-Invasive Monitoring of Brain Tissue Temperature by Near-Infrared*
- 30 *Spectroscopy*, Ph.D. thesis, University College London.
- 31 Kennedy, P. *et al.* [2000] “Direct control of a computer from the human central nervous
- 32 system,” *IEEE Transactions on Rehabilitation Engineering* **8**(2), 198–202.
- 33 Kim, J., White, K. and Winfree, E. [2006] “Construction of an *in vitro* bistable circuit from
- 34 synthetic transcriptional switches,” *Molecular Systems Biology* **2**(68), 1–12, doi: 10.1038/
- 35 msb4100099.
- 36 Kircher, M. *et al.* [2003] “A multimodal nanoparticle for preoperative magnetic resonance
- 37 imaging and intraoperative optical brain tumor delineation,” *Cancer Research* **63**,
- 38 8122–8125.
- 39 Knott, G. *et al.* [2008] “Serial section scanning electron microscopy of adult brain tissue using
- 40 focused ion beam milling,” *Journal of Neuroscience* **28**(12), 2959–2964.
- 41 Koene, R. [2011] “Substrate-independent minds: Pattern survival agrees with universal
- 42 darwinism,” in *Future of Humanity Institute Winter Intelligence Conference* (Oxford, UK),
- [http://www.carboncopies.org/future-of-humanity-institute-](http://www.carboncopies.org/future-of-humanity-institute-winter-intelligence-conference-oxford-uk-14-17-january-2011)
- [winter-intelligence-conference-oxford-uk-14-17-january-2011](http://www.carboncopies.org/future-of-humanity-institute-winter-intelligence-conference-oxford-uk-14-17-january-2011).
- Koene, R. [2012] “Fundamentals of whole brain emulation: State, transition and update
- representations,” *International Journal of Machine Consciousness*.
- Kondandaramaiah, S., Boyden, E. and Forest, C. [2011] “Automated patch clamping of
- neurons in the mammalian brain *in vivo*,” in *2011 Neuroscience Meeting Planner* (Society
- for Neuroscience, Washington, DC, 2011), [Online.], p. Program No. 305.03.

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- 1 Kralj, J. *et al.* [2011] “Optical recording of action potentials in mammalian neurons using a
- 2 microbial rhodopsin,” *Nature Methods*, doi: 10.1038/nmeth.1782.
- 3 McCormick, B. and Mayerich, D. [2004] “Three-dimensional imaging using knife-edge scanning
- 4 microscopy,” *Proc. Microscopy and Micro-Analysis Conference 2004*, Savannah, GA,
- 5 pp. 1466–1467.
- 6 Oyibo, H. *et al.* [2011] “Neural connectivity as a DNA sequencing problem *in vivo*,” in *Program*
- 7 *no. 617.25. 2011 Neuroscience Meeting Planner* (Society for Neuroscience, Washington,
- 8 DC), p. Online.
- 9 Raper, J. and Mason, C. [2010] “Cellular strategies of axonal pathfinding,” *Cold Spring Harbor*
- 10 *Perspectives in Biology* **2**(9), a001933.
- 11 Reddy, G. *et al.* [2006] “Vascular targeted nanoparticles for imaging and treatment of brain
- 12 tumors,” *Clinical Cancer Research* **12**(22), 6677–6686.
- 13 Reed, M. *et al.* [1988] “Observation of discrete electronic states in a zero-dimensional semi-
- 14 conductor nanostructure,” *Physics Review Letters* **60**(6), 535–537, doi: 10.1103/Phys-
- 15 RevLett.60.535.
- 16 Rothermund, P. [2006] “Folding DNA to create nanoscale shapes and patterns,” *Nature*
- 17 **440**(7082), 297–302.
- 18 Sakai, H. [1992] “White-noise analysis in neurophysiology,” *Physiological Reviews* **72**(2),
- 19 491–505.
- 20 Seung, S. [2012] *Connectome: How the Brain’s Wiring Makes Us Who We Are* (Houghton
- 21 Mifflin Harcourt).
- 22 Sierra, D., Weir, N. and Jones, J. [2005] “A review of research in the field of nanorobotics,”
- 23 Tech. Rep. SAND2005-6808,1-50, U.S. Department of Energy — Office of Scientific and
- 24 Technical Information Oak Ridge, TN.
- 25 Thyagarajan, A. and Ting, A. [2010] “Imaging activity-dependent regulation of neuroligin-
- 26 neuroligin interactions using *trans*-synaptic enzymatic biotinylation,” *Cell* **143**, 456–469.
- 27 Tian, J. *et al.* [2004] “Accurate multiplex gene synthesis from programmable DNA chips,”
- 28 *Nature* **432**(23/30), 1050–1054.
- 29 Walling, M., Novak, J. and Shepard, J. [2009] “Quantum dots for live cell and *in vivo*
- 30 imaging,” *International Journal of Molecular Sciences* **10**(2), 441–491, doi: 10.3390/
- 31 ijms10020441.
- 32 Watanabe, H. *et al.* [2009] “Intravascular neural interface with nanowire electrode,” *Elec-*
- 33 *tronics and Communications in Japan* **92**(7), 29–37.
- 34 Yamamoto, Y. and Shinohara, K. [2002] “Application of X-ray microscopy in analysis of living
- 35 hydrated cells,” *The Anatomical Record* **269**(5), 217–223.
- 36 Zador, A. [2011] “Sequencing the connectome: A fundamentally new way of determining the
- 37 brains wiring diagram,” Tech. rep., Project Proposal, Paul G. Allen Foundation Awards
- 38 Grants.
- 39 Zorzos, A. *et al.* [2011a] “A novel 3-d microfabricated lightguide array for optogenetic control
- 40 of distributed neural circuits in the mammalian brain,” in *2011 Neuroscience Meeting*
- 41 *Planner* (Society for Neuroscience, Washington, DC, 2011), [Online.], p. Program No.
- 42 305.06.
- Zorzos, A. *et al.* [2011b] “High channel-count silicon neural recording probes for 3d charac-
- terization of optogenetically modulated neural dynamics,” in *2011 Neuroscience Meeting*
- Planner* (Society for Neuroscience, Washington, DC, 2011), [Online.], p. Program
- No. 305.01.